Machine perception

Practicum

Assistants

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Time slots

- 5 available time slots (urnik.fri.uni-lj.si)
  - All time slots will be held **in person**
  - Attendance is mandatory, but only for **defense**
- You must attend the chosen time slot for the duration of the semester

Practicum grading

- 6 assignments (one every two weeks)
  - Mandatory tasks (75 points)
  - Optional tasks (25 or more points)
  - Modifiers
    - Poor defense (-N points)
    - Late submission or late defense (score penalty x0.75)
    - Joker week (score penalty x0.75)
- Final score: total points / 6, need 50 to pass, max 100

Assignments

- Assignments in Python
  - Local install or Google Colab
  - Jupyter is ok (with a note related to debugging)
- Each assignment must be defended with the assistant
- Assignments must be submitted to e-classroom before the defense
  - *.py source files in a zip file
- The deadline is Monday morning of the defense week
- Please write and debug your own code

Consultations

- Scheduled every other week
- Same time slots
- The assistant will be available for questions
- Common problems or bugs will be addressed
- Student interaction is required (i.e. you must ask specific questions)

Defense

- 5 minutes per person (be prepared!)
- Prepare the presentation of your solutions beforehand
  - i.e., run slow computations before the presentation
- The assistant will pose some questions about your solution
- The questions might be open-ended or theoretical as well

Exercises at a glance

Exercise 1 — Basic image processing, histograms

Exercise 2 — Recognition with histograms, convolution

Exercise 3 — Edge detection, Hough transform

Exercise 4 — Homography, feature matching

Exercise 5 — Epipolar geometry and triangulation

Exercise 6 — Image subspaces

Joker week — mid-January

Week overview

Week #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Available		1		2		3		4		5		6	holidays	holidays	6	
Defense				1		2		3		4		5				J
Late defense					1		2		3		4				5	6